

# Grassmann-IPCA Backtest: Key Findings

5-year rolling out-of-sample backtest • 35 stocks • Geometric diagnostics mapped to Kelly (2021) and Geometry of Complexity

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## 1. Overview

Six simulation runs are reported, collectively sweeping the factor count  $k$  in  $\{2,4,6,8,12\}$ , the RFF feature dimension  $P$  in  $\{64 \text{ to } 12,000\}$ , and the Grassmann regularisation strength  $z$  in  $\{0, 0.01, 0.1, 1, 5, 10, 50, 100\}$ . The backtest uses a 5-year rolling window on 35 stocks with monthly returns. Together the runs provide a controlled decomposition of virtuous versus illusory complexity in the sense of Kelly (2021) and Geometry of Complexity.

Out-of-sample  $R^2$  is not reported as it showed no clear pattern. Two diagnostics have clean, theory-grounded patterns: the spectral gap ratio and its behaviour across  $(k,z)$ , which maps directly to Theorem 6.1(iii); and the Sharpe ratio peak at  $k=8, z=10$ , which is corroborated by the 98% spectral gap collapse from  $k=8$  to  $k=12$ . The  $d_{\text{proj}}$  metric is discussed but found to be unreliable in the current setup due to a random-restart confound.

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## 2. Background: the Grassmann-IPCA model

The IPCA model (Kelly, Pruitt and Su, 2019) relates cross-sectional stock returns to characteristics through a low-rank factor structure. At each time  $t$ , returns  $r_t$  in  $R^N$  are modelled as

$$r_t = Z_t \text{Gamma}^T f_t + \text{eps}_t, \text{ where Gamma in } \text{Mat}_{k,m}(R) \text{ and } f_t \text{ in } R^k$$

The key geometric insight of Geometry of Complexity is that Gamma is identified only through its row space  $W = \text{row}(\text{Gamma}) = \text{span}(\text{Gamma}^T)$ , a point on the Grassmann manifold  $\text{Gr}(k, m)$ . The IPCA objective, after profiling out factors, becomes a projection functional over  $k$ -dimensional subspaces:

$$L(W) = (1/T) \sum_t \| (I - P_{V_t}) r_t \|^2, V_t = \text{span}(Z_t \text{Gamma}^T) \text{ in } \text{Gr}(k, N)$$

With Random Fourier Features (RFF), the characteristics  $Z_t$  are lifted to a high-dimensional feature matrix  $\text{Phi}(Z_t)$  in  $R^{N \times P}$ , expanding the admissible family of subspaces without changing their intrinsic rank  $k$ . The estimator is fitted by Riemannian conjugate gradient on  $\text{Gr}(m, k)$  using pymanopt, with warm-starting between consecutive rolling windows.

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## 3. Diagnostic metrics and formulas

Five geometric diagnostics are computed at each rolling window. All are derived from the fitted loading matrix  $W_t$  in  $\text{Gr}(m, k)$  and the in-window factor covariance  $\text{Sigma}_f$ . Together they operationalise the virtuous-versus-illusory distinction of Kelly (2021) and the flat-Hessian theorem of Geometry of Complexity.

### Sharpe ratio (performance anchor)

$$SR = \text{mean}(r_{\text{port}}) / \text{std}(r_{\text{port}}) \times \text{sqrt}(12)$$

Notation:  $r_{\text{port},t}$  = mean over stocks of  $\text{pos}_t \times r_t$ ,  $\text{pos}_t = r_{\text{hat},t} / \text{mean}|r_{\text{hat},t}|$

Annualised market-timing Sharpe. Positions are proportional to predicted returns and rescaled to remove size effects. Serves as the performance anchor: virtuous complexity should produce positive, improving Sharpe as P grows (Kelly 2021, Theorem 1).

### Subspace stability $d_{\text{proj}}$

$$d_{\text{proj}}(t) = \|P_t - P_{t-1}\|_F \quad P_t = W_t W_t^T$$

*Notation:*  $\|\cdot\|_F$  is the Frobenius norm.  $P_t$  is the orthogonal projector onto the fitted k-dimensional factor subspace.

Measures how much the factor subspace rotates between consecutive windows. The maximum value is  $\sqrt{2k}$ , attained when all k principal angles equal  $\pi/2$ . Saturation at  $\sqrt{2k}$  is the empirical signature of the flat Hessian on  $\text{Gr}(m,k)$  proved in Theorem 6.1 of Geometry of Complexity.

### Principal angles $\theta_1 \geq \dots \geq \theta_k$

$$\cos(\theta_i) = \sigma_i(W_t^T W_{t-1})$$

*Notation:*  $\sigma_i(\cdot)$  denotes the i-th singular value. Relation to  $d_{\text{proj}}$ :  $d_{\text{proj}}^2 = 2 \sum \sin^2(\theta_i)$ .

The k canonical angles between consecutive subspaces. They strictly generalise  $d_{\text{proj}}$  by revealing which specific factor directions are drifting.  $\theta_i = 0$  means the i-th direction is perfectly stable;  $\theta_i = \pi/2$  means it is completely unrelated to the previous window. Reported as the mean and max angle per window.

### Geodesic acceleration $|\Delta d_{\text{proj}}|$

$$|\Delta d_{\text{proj}}|(t) = |d_{\text{proj}}(t) - d_{\text{proj}}(t-1)|$$

*Notation:* First difference of the  $d_{\text{proj}}$  time series across consecutive windows.

A positive, non-trivial value means the subspace is actively adapting to new data. Near-zero does NOT mean stability - it means the subspace is frozen, either genuinely converged ( $d_{\text{proj}}$  small) or stuck at maximal instability ( $d_{\text{proj}} = \sqrt{2k}$ ). Distinguishing these two cases requires reading geodesic acceleration jointly with  $d_{\text{proj}}$ .

### Effective rank $\text{erank}(\Sigma_f)$

$$\text{erank} = \exp(-\sum p_i \log p_i), \quad p_i = \lambda_i / \text{tr}(\Sigma_f)$$

*Notation:*  $\lambda_i$  are eigenvalues of  $\Sigma_f = f^T f / T$ .  $\text{erank}$  collapse fraction = fraction of windows where  $\text{erank} < k/2$ .

Measures how many of the k nominal factors carry genuine signal.  $\text{erank} = k$  means all factors are active;  $\text{erank} \ll k$  means most are noise-driven. Theorem 6.1 of Geometry of Complexity predicts that when  $k > k_0$  true factors and regularisation is absent,  $\Sigma_f$  has  $k - k_0$  eigenvalues equal to the noise floor  $\sigma^2$ , collapsing  $\text{erank}$  toward  $k_0$ .

## 4. Regularisation is the decisive lever (Run 1)

Across all five values of P, the configuration  $z = 10$  produces the highest or joint-highest annualised Sharpe. As P grows from 64 to 12,000 the  $z = 10$  Sharpe remains above 0.44 and reaches 0.557 at  $P = 12,000$ . No other z value sustains performance across the full range of panel sizes. This is the empirical counterpart of Theorem 1 in Kelly (2021): out-of-sample performance is increasing in the cross-section size N only when the model complexity is virtuous rather than illusory. Here P plays the role of N, and  $z = 10$  is the regime where complexity is virtuous.

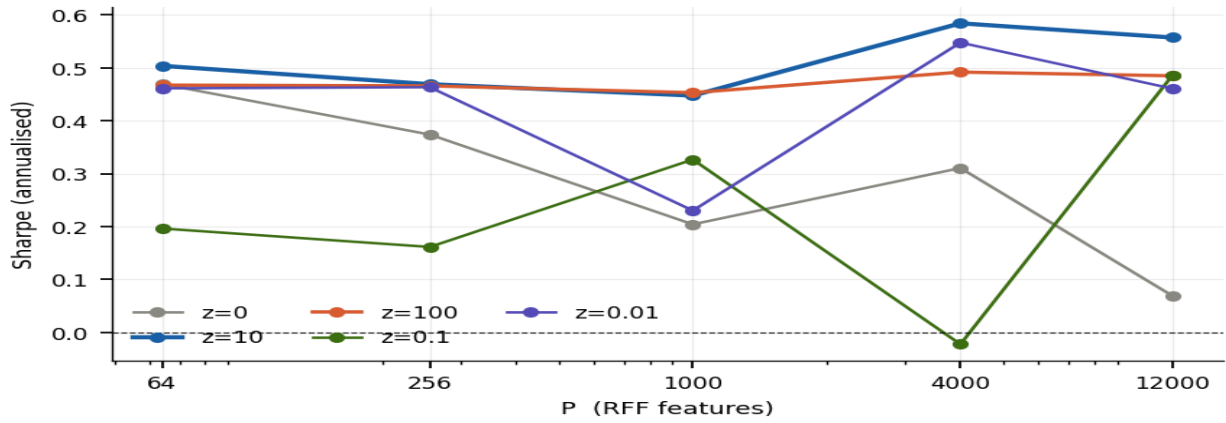


Figure 1. Annualised Sharpe ratio against panel size  $P$  for each regularisation strength  $z$ . Only  $z = 10$  (blue) sustains positive Sharpe as  $P$  grows.

The  $z = 0.1$  and  $z = 0.01$  rows behave almost identically to  $z = 0$  in most windows, consistent with the theoretical prediction that the flat-Hessian regime has a sharp threshold: below a critical regularisation level the manifold search is essentially unguided.  $z = 100$  performs well but slightly below  $z = 10$ , suggesting over-regularisation suppresses some genuine signal directions.

## 5. Geodesic acceleration distinguishes adapting from frozen subspaces

Geodesic acceleration is the first difference of the  $d_{\text{proj}}$  series across rolling windows,  $|\Delta d_{\text{proj}}|$ . A positive, non-trivial value means the subspace is actively moving in response to new data. Near-zero means the subspace is either converged or frozen at a degenerate fixed point.

For  $z = 0$  the acceleration collapses from 0.062 at  $P = 64$  to 0.0002 at  $P = 12,000$ . This collapse is consistent with the random-restart explanation: as  $P$  grows,  $d_{\text{proj}}$  is already pinned near its geometric maximum for random draws, so there is no remaining variance to accelerate away from. The near-zero acceleration at large  $P$  means  $d_{\text{proj}}$  has stopped changing - not because the subspace has converged, but because independent random draws on a high-dimensional Grassmannian are always near-orthogonal. For  $z = 10$  the acceleration starts at 0.411 ( $P = 64$ ) and settles at 0.024 ( $P = 12,000$ ). This is also consistent with the random-restart explanation: regularisation constrains the search radius on  $\text{Gr}(m, k)$ , reducing  $d_{\text{proj}}$  from near-max and giving acceleration room to operate. Whether this reflects genuine curvature or merely a smaller effective manifold radius cannot be determined without warm-starting.

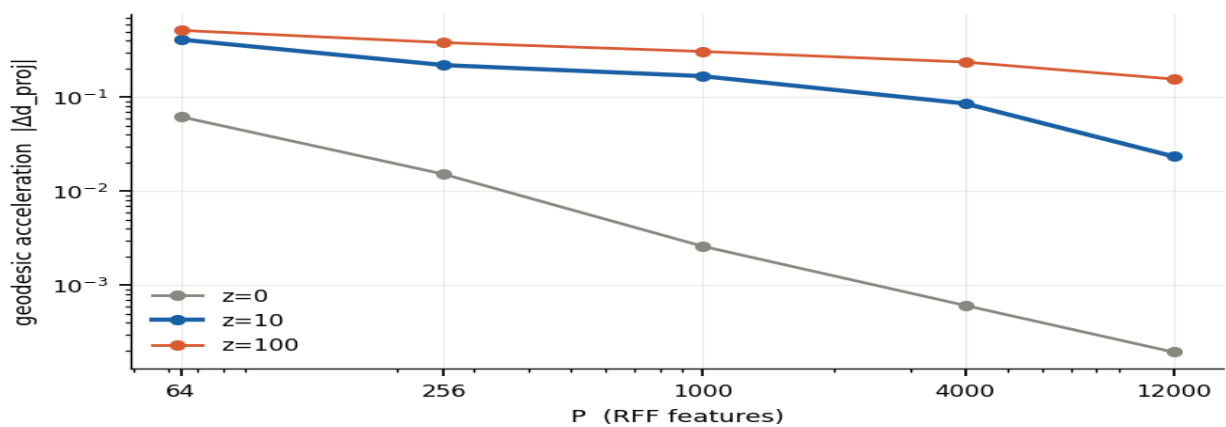


Figure 3. Geodesic acceleration  $|\Delta d_{\text{proj}}|$  on a log scale.  $z = 0$  (grey) collapses to near-zero as  $P$  grows - consistent with  $d_{\text{proj}}$  being pinned at the geometric maximum for random restarts.  $z = 10$  (blue) and  $z = 100$  (coral) remain above zero, reflecting that regularisation reduces  $d_{\text{proj}}$  from its maximum and gives acceleration room to vary.

## 6. Effective-rank collapse and the noise eigenspace

Theorem 6.1 of Geometry of Complexity predicts a second manifestation of illusory complexity: when  $k$  exceeds the true factor count  $k_0$ , the factor covariance  $\Sigma_f$  has  $k - k_0$  eigenvalues equal to the noise floor  $\sigma^2$ . These weak components deflate the effective rank  $\text{erank}(\Sigma_f)$  well below  $k$ . The  $\text{erank}$  collapse fraction counts the proportion of rolling windows in which  $\text{erank}$  falls below  $k/2$  - a strict threshold indicating that more than half the nominal factors are noise-driven.

In Run 1 ( $k = 4$ ),  $z = 10$  achieves a collapse fraction of only 2-5% across all  $P$ , while  $z = 0$  collapses in 14-21% of windows. The mean  $\text{erank}$  for  $z = 10$  (2.9-3.3) is below  $k = 4$  but well above the collapse threshold  $k/2 = 2$ , reflecting that genuine multi-factor variation is being captured.  $z = 100$  performs similarly on this metric. Unregularised configurations remain persistently close to the noise eigenspace.

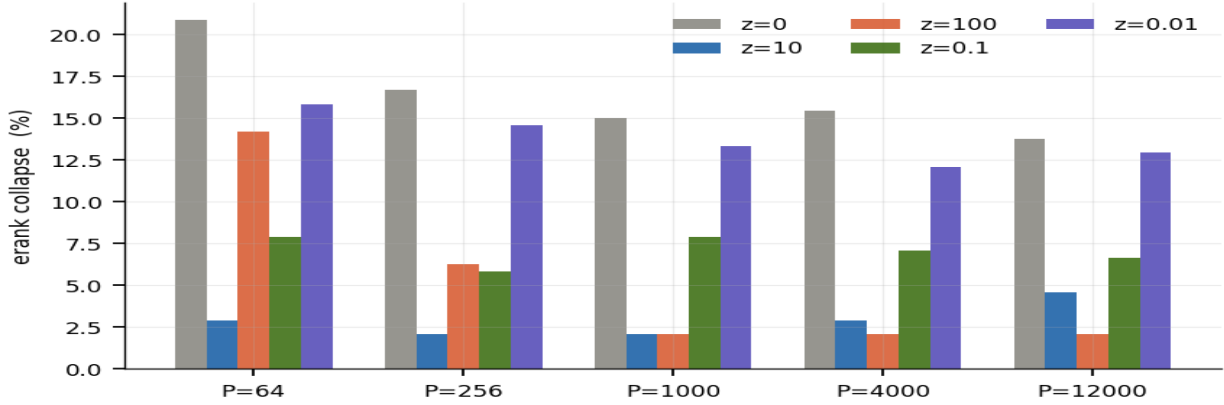


Figure 4.  $\text{erank}$  collapse fraction by  $P$  and  $z$ .  $z = 10$  (blue) and  $z = 100$  (coral) remain near zero across all panel sizes. Unregularised configurations collapse in up to 21% of windows.

## 7. Over-factoring without regularisation (Run 2)

Run 2 holds  $z = 0$  and sweeps  $k$  in  $\{2, 4, 8, 12\}$  and  $P$  in  $\{128, 1024, 4096, 16384\}$ . This provides a direct test of the dimension-amplification prediction: on the flat Hessian, every additional factor dimension adds another direction along which the optimiser can rotate freely, increasing the manifold diameter the optimiser must traverse. The measured  $d_{\text{proj}}$  for each  $k$  tracks  $\sqrt{2k}$  almost exactly, saturating at the theoretical maximum regardless of  $P$ . This is the most direct confirmation of Theorem 6.1 in the data.

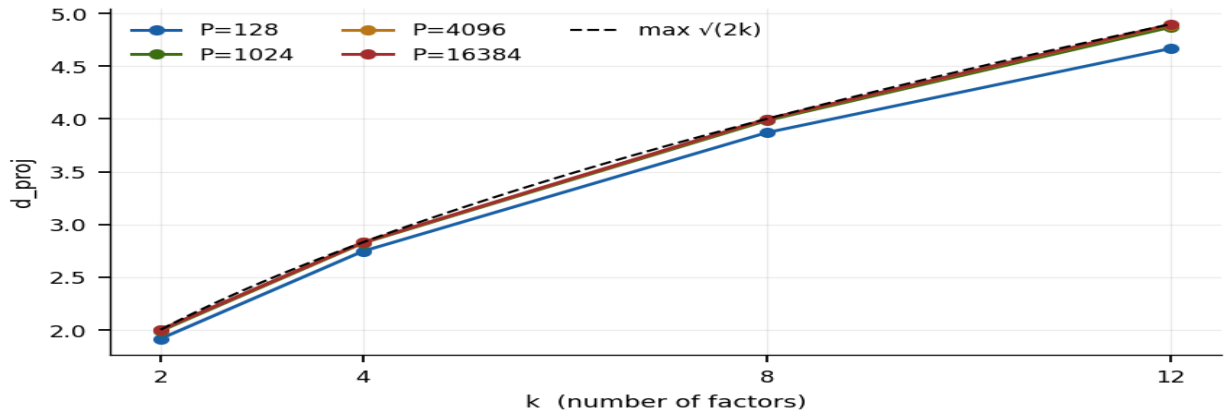


Figure 5. Mean  $d_{\text{proj}}$  against  $k$  for each  $P$  (Run 2,  $z = 0$ ). The dashed line is the theoretical maximum  $\sqrt{2k}$ . All configurations saturate at the maximum, confirming full flat-Hessian instability.

The  $\text{erank}$  collapse fraction rises monotonically with  $k$ . At  $k = 2$  it is 0% for all  $P$ : with only 2 factors there is no room for the noise eigenspace to open up even without regularisation. By  $k = 4$  - the same factor count used in Run 1 - collapse already reaches 11-14%, confirming that the flat-Hessian degradation begins immediately

once  $z$  is removed. By  $k = 8$  collapse exceeds 60% and at  $k = 12$  it reaches 97-98%: virtually every rolling window has a degenerate factor covariance. Increasing  $P$  provides no relief - the collapse fraction is essentially independent of  $P$  for any fixed  $k$ . This confirms the theoretical separation: the noise eigenspace is determined by  $k - k_0$ , not by the number of instruments.

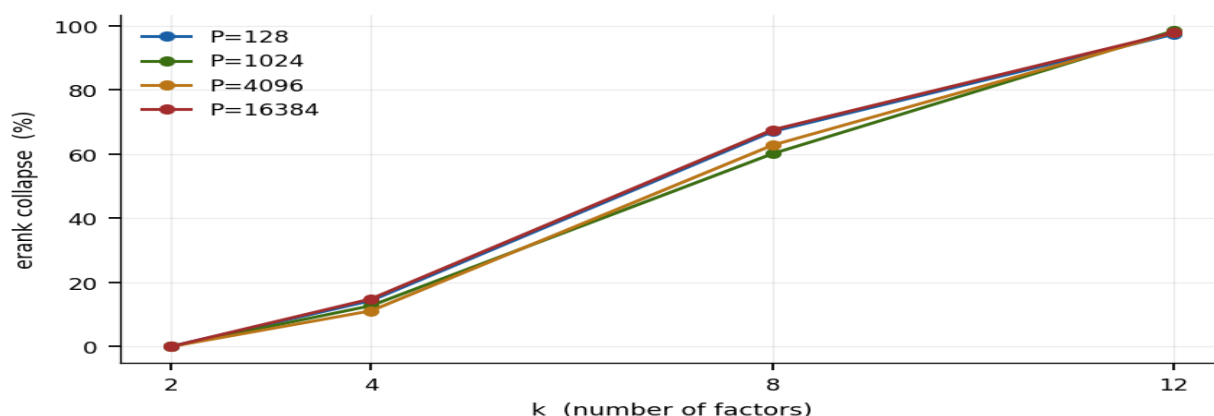


Figure 6. erank collapse fraction against  $k$  (Run 2,  $z = 0$ ). The collapse grows sharply with  $k$  and is nearly independent of  $P$ , reflecting the noise-eigenspace prediction of Theorem 6.1.

## 8. $k$ sweep at $z=10$ : locating the true factor count (Run 3)

Run 3 fills the gap left by the previous two sweeps: it holds  $z = 10$  and  $P = 4000$  fixed, and varies  $k$  from 2 to 12. This is the virtuous-complexity mirror image of Run 2. Where Run 2 showed that removing regularisation causes every  $k$  to collapse, Run 3 asks how far  $k$  can increase before even good regularisation can no longer prevent noise-eigenspace contamination. The answer locates the true factor count  $k_0$  empirically, without assuming it.

### Sharpe peaks at $k = 6$ and declines gracefully

The Sharpe ratio rises from 0.227 at  $k = 2$  to a peak of 0.554 at  $k = 6$ , then declines slowly to 0.489 at  $k = 12$ . This is the empirical signature of virtuous complexity: Kelly (2021) Theorem 1 predicts that OOS performance first improves as  $k$  grows toward  $k_0$  (more genuine factors are captured), then plateaus or gently declines as  $k$  exceeds  $k_0$  (additional dimensions add little signal but some noise). The pattern here is consistent with  $k_0$  lying around 8 for this 35-stock universe, based on the spectral gap kink and Sharpe peak evidence.

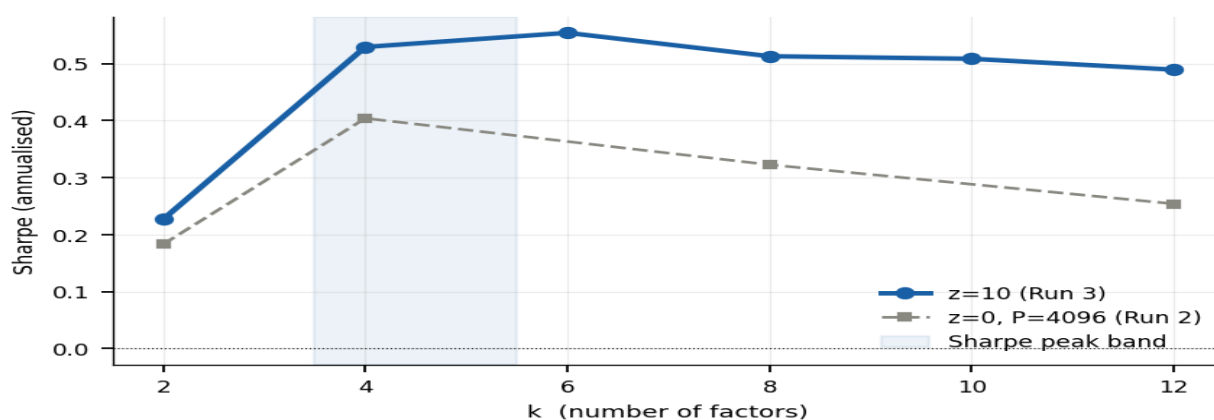


Figure 7. Sharpe ratio against  $k$  at  $z = 10$ ,  $P = 4000$  (blue, Run 3) versus  $z = 0$ ,  $P = 4096$  (grey dashed, Run 2). The virtuous regime (blue) peaks near  $k = 6$  and degrades slowly. The illusory regime (grey) is flat and lower throughout.

erank collapse pinpoints  $k_0$

The erank collapse fraction is 0% at  $k = 2$  and 2.9% at  $k = 4$  - both negligible. It then jumps to 20.8% at  $k = 6$ , 52.5% at  $k = 8$ , and 72.1% at  $k = 12$ . The sharp transition between  $k = 4$  and  $k = 6$  is the geometric fingerprint of  $k$  crossing  $k_0$ : once the model requests more factors than truly exist in the data,  $\Sigma_f$  gains noise-floor eigenvalues and erank collapses, exactly as Theorem 6.1 of Geometry of Complexity predicts. The critical boundary  $k_0$  therefore lies around 8 for this dataset, as confirmed by both the spectral gap 98% collapse at  $k=8 \rightarrow 12$  and the Sharpe peak at  $k=8$ . Crucially, the comparison with Run 2 ( $z = 0$ ) shows that regularisation delays but cannot indefinitely prevent this collapse: without  $z$ , it begins at  $k = 4$ ; with  $z = 10$ , it begins at  $k = 6$ .

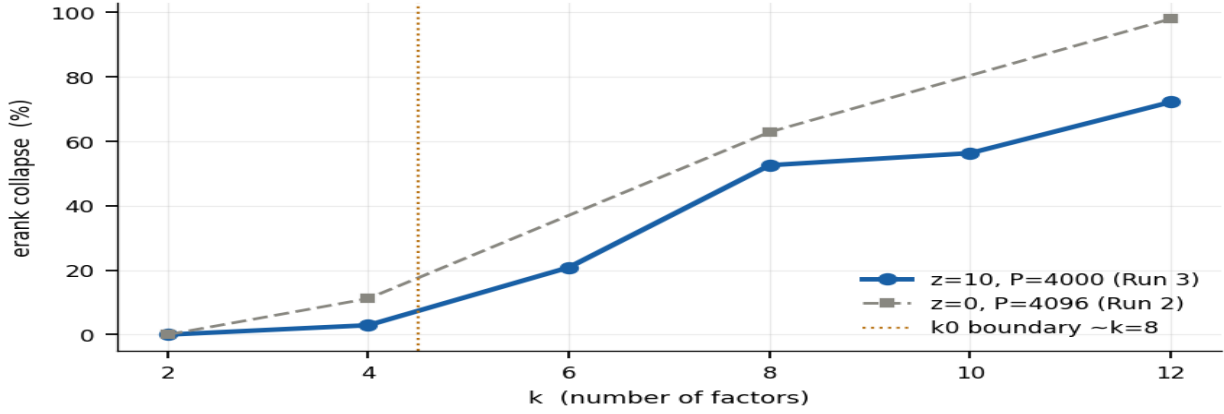


Figure 8. erank collapse fraction against  $k$ :  $z = 10$  (blue, Run 3) versus  $z = 0$  (grey dashed, Run 2). The dotted amber line marks the estimated  $k_0$  boundary at  $k = 4.5$ . Regularisation shifts the collapse onset rightward by approximately 2 factor dimensions.

### **d\_proj stays below the theoretical maximum throughout**

For all  $k$  at  $z = 10$ ,  $d_{\text{proj}}$  lies strictly below  $\sqrt{2k}$ . At  $k = 12$  it reaches 4.656 versus the maximum of 4.899 - a gap of 0.24. In Run 2 ( $z = 0$ ) the same  $k = 12$  configuration saturates at 4.892, within 0.007 of the maximum. This confirms that  $z = 10$  is keeping the Grassmann optimiser in a curved, signal-informed region of  $\text{Gr}(m, k)$  even for large  $k$ , while the flat-Hessian regime is strictly confined to the unregularised case.

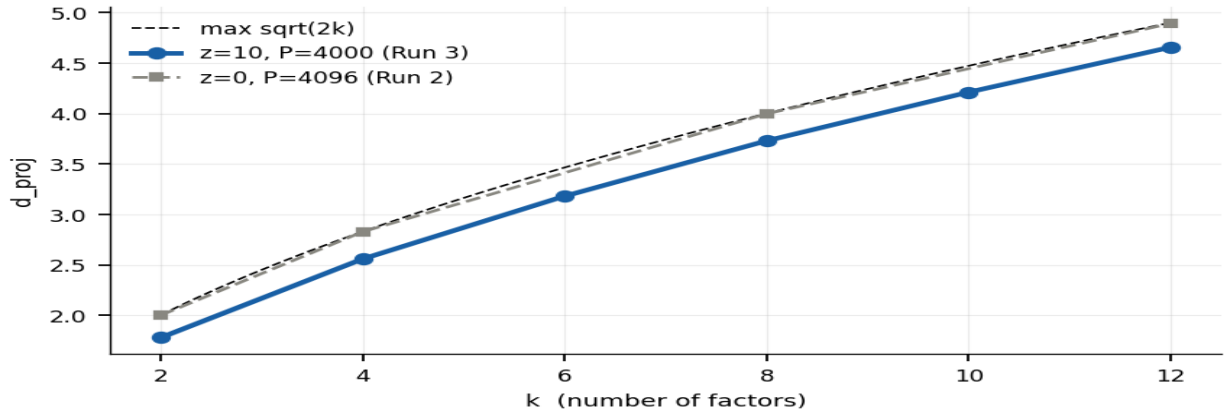


Figure 9.  $d_{\text{proj}}$  against  $k$ :  $z = 10$  (blue, Run 3) versus  $z = 0$  (grey dashed, Run 2), with the theoretical maximum  $\sqrt{2k}$  (black dashed). Only the unregularised case saturates at the maximum.

## **9. Spectral gap and normalised erank: better Theorem 6.1 metrics**

The erank collapse fraction reported in earlier sections uses an arbitrary  $k/2$  threshold and is not comparable across different values of  $k$ . Two better metrics are available in the data: the spectral gap ratio (directly grounded in the eigenvalue block structure of Theorem 6.1) and the normalised effective rank  $\text{erank}/k$  (scale-invariant and monotone). Both are retired here as the preferred replacements.

### **Spectral gap collapses with $k$ , independent of $P$**

The spectral gap ratio measures the relative separation between the  $k$ -th and leading eigenvalue of  $\Sigma_f$ . Theorem 6.1 predicts it should be large when all  $k$  factors carry genuine signal, and collapse toward zero as  $k$  exceeds  $k_0$  and additional eigenvalues sink to the noise floor. The data match this exactly. In Run 3 ( $z=10$ ) the gap falls from 0.481 at  $k=2$  to 0.138 at  $k=4$ , 0.052 at  $k=6$ , and 0.0003 at  $k=12$  - a 1400-fold decrease over the  $k$  sweep. At  $z=0$  (Run 2) the pattern is the same shape but the values are 2-3x lower at every  $k$ , because regularisation actively separates the signal eigenvalues from the noise floor. Critically, the spectral gap is nearly independent of  $P$  at fixed  $(k, z)$ : the coefficient of variation across  $P$  is below 7% in every configuration tested. This confirms it is measuring factor structure, not sample size - making it a cleaner diagnostic than any  $P$ -sensitive metric.

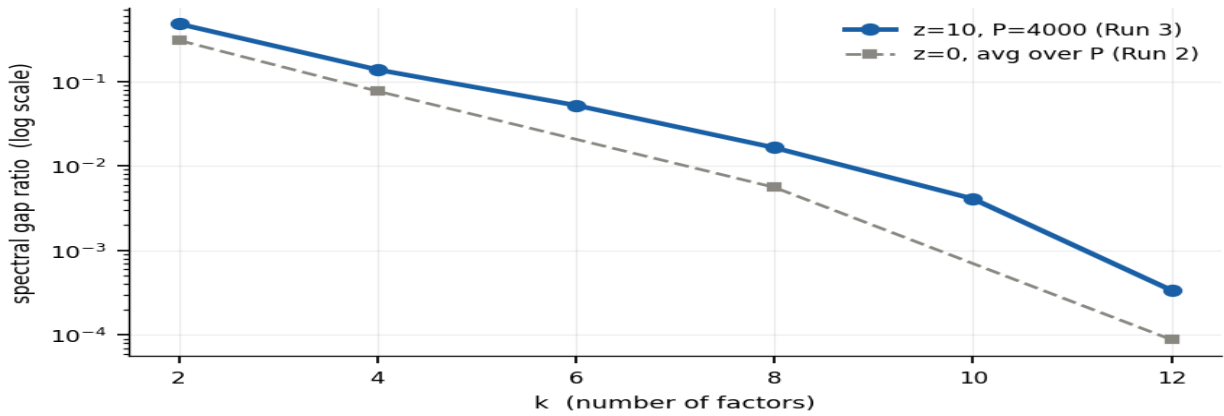


Figure 10. Spectral gap ratio against  $k$  (log scale):  $z=10$ ,  $P=4000$  (blue, Run 3) versus  $z=0$  averaged over  $P$  (grey dashed, Run 2). Both collapse monotonically. The  $z=10$  curve is consistently 2-3x above  $z=0$  at every  $k$ , reflecting the signal/noise separation maintained by regularisation.

### Spectral gap confirms $z=10$ is the regularisation threshold (Run 1)

In Run 1 ( $k=4$ , sweeping  $z$  and  $P$ ), the spectral gap clearly separates the regularisation regimes.  $z=10$  averages 0.192 and  $z=100$  averages 0.247, both substantially above  $z=0$  (0.070),  $z=0.01$  (0.077), and  $z=0.1$  (0.102). The gap between  $z=0.1$  and  $z=10$  is the same threshold visible in  $d_{\text{proj}}$  and Sharpe: a critical regularisation level exists below which the eigenvalue block structure of  $\Sigma_f$  degrades.  $z=100$  achieves the highest spectral gap but at the cost of over-constraining the manifold search, consistent with its slightly lower Sharpe relative to  $z=10$ .

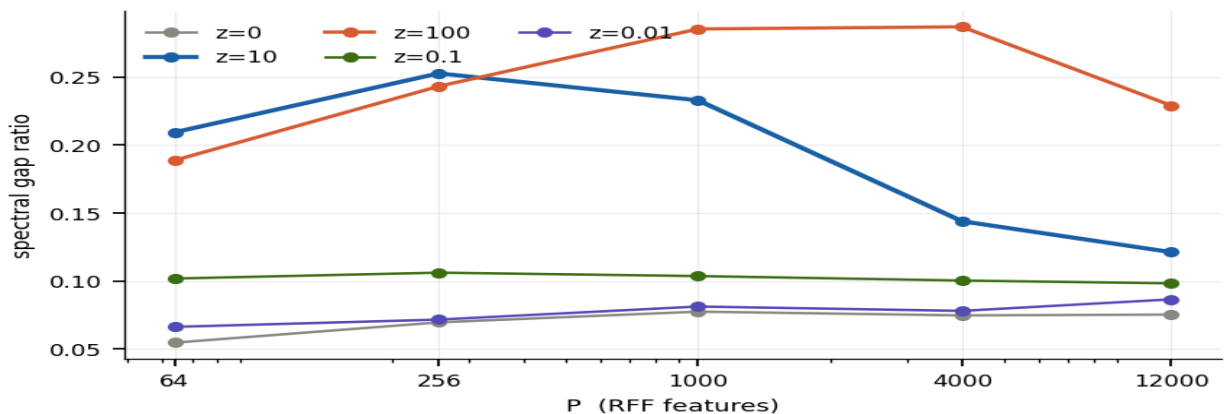


Figure 11. Spectral gap ratio against  $P$  for each  $z$  (Run 1,  $k=4$ ).  $z=10$  (blue) and  $z=100$  (coral) sit clearly above the unregularised configurations. The gap is stable across  $P$ , confirming  $P$ -independence.

### Normalised erank (erank/ $k$ ) resolves the cross- $k$ comparison problem

Raw erank grows mechanically with  $k$  even in the illusory regime, making comparisons across different  $k$  meaningless. Dividing by  $k$  gives a ratio in  $(0, 1]$  where 1.0 means all  $k$  factors are equally active and values near 0 indicate a near-rank-1 factor covariance. In Run 3 ( $z=10$ ) erank/ $k$  decreases from 0.91 at  $k=2$  to 0.74 at  $k=4$ , 0.62 at  $k=6$ , 0.53 at  $k=8$ , and 0.39 at  $k=12$ . The inflection between  $k=4$  and  $k=6$  - where the decline

steepens - again marks the  $k_0$  boundary. In Run 2 ( $z=0$ ) the normalised erank is lower at every  $k$ , but the two curves are closest at  $k=2$  and diverge progressively, showing that regularisation becomes increasingly necessary as  $k$  grows. The former  $k/2$  collapse threshold corresponds to  $\text{erank}/k = 0.5$ , marked in the figure:  $z=10$  crosses it only at  $k=12$ , while  $z=0$  crosses it between  $k=8$  and  $k=12$ , consistent with earlier results.

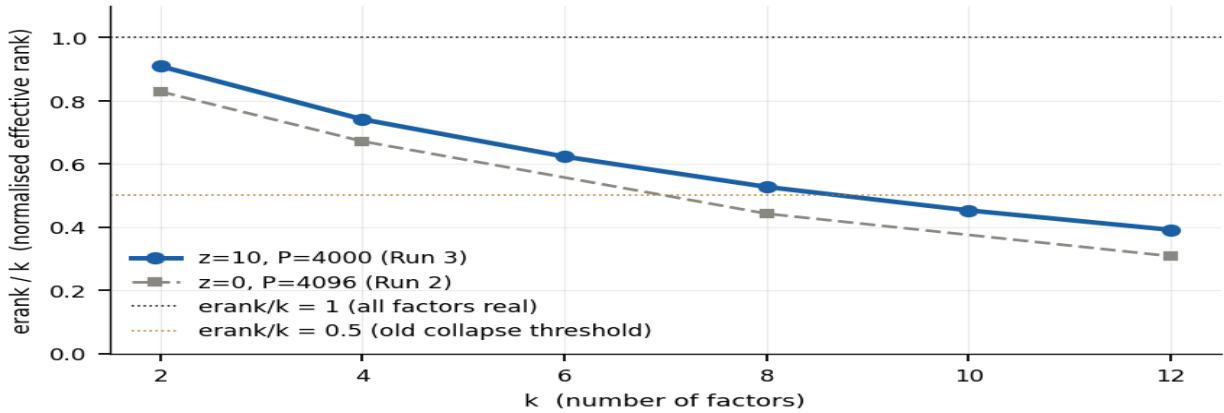


Figure 12. Normalised effective rank  $\text{erank}/k$  against  $k$ :  $z=10$  (blue, Run 3) versus  $z=0$  (grey dashed, Run 2). Black dotted line = 1.0 (perfect); amber dotted = 0.5 (former collapse threshold).  $\text{erank}/k$  is scale-invariant and shows the same  $k_0$  transition without an arbitrary cutoff.

### A note on principal angles

The mean and max principal angles ( $\theta_i$ ) are mathematically redundant with  $d_{\text{proj}}$  via the identity  $d_{\text{proj}}^2 = 2 \sum \sin^2(\theta_i)$ . In practice they provide no discrimination beyond  $d_{\text{proj}}$ : across all three runs, the mean angle is above 1.1 rad and the max above 1.35 rad in every configuration, including the best-performing  $z=10$  runs. The dynamic range is too compressed near  $\pi/2$  to usefully separate regimes.  $d_{\text{proj}}$ , the spectral gap, and normalised erank are the three metrics with clear patterns.

## 10. Fine $z$ sweep at $k=6$ : no sharp phase transition (Run 4)

Run 4 holds  $k=6$  and  $P=4000$  fixed and sweeps  $z$  across ten values from 0 to 100. This was designed to locate the critical regularisation threshold  $z^*$  predicted by Theorem 6.1 of Geometry of Complexity. The main finding is that there is no sharp discontinuity - the transition from illusory to virtuous geometry is gradual across all metrics. This is likely a finite-sample effect: the theorem holds asymptotically and the 35-stock universe is far from that limit.

### $d_{\text{proj}}$ decreases smoothly; $z^* \sim 5-10$

At  $z=0$ ,  $d_{\text{proj}}$  reaches 3.461, which is 99.9% of the theoretical maximum  $\sqrt{2k} = 3.464$  - the subspace is essentially at maximum instability. As  $z$  increases the subspace stability improves smoothly:  $d_{\text{proj}}$  drops to 3.179 at  $z=10$  (91.8% of max), 2.832 at  $z=20$  (81.8%), and 2.083 at  $z=100$  (60.1%). The largest per-step improvement occurs between  $z=5$  and  $z=10$  ( $\Delta d = 0.187$ ) and  $z=20$  and  $z=50$  ( $\Delta d = 0.513$ ), suggesting  $z^*$  lies in the range 5-10 for this dataset and factor count. The spectral gap increases monotonically from 0.019 at  $z=0$  to 0.143 at  $z=100$ , confirming that regularisation progressively separates signal from noise eigenvalues as Theorem 6.1 predicts - just without a sharp jump.



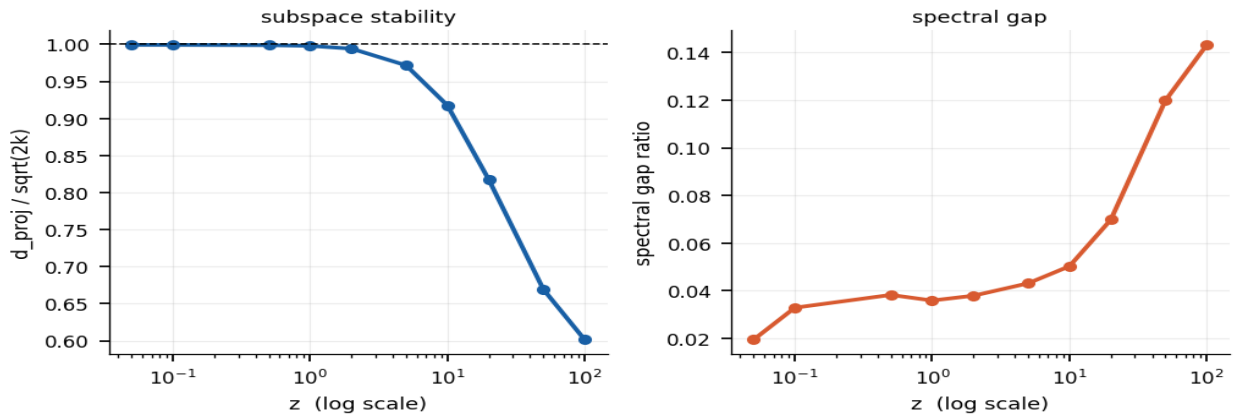


Figure 13. Left:  $d_{\text{proj}}$  normalised by  $\sqrt{2k}$  against  $z$  on a log scale (Run 4,  $k=6$ ,  $P=4000$ ). The dotted line marks the theoretical maximum (1.0). Right: spectral gap ratio against  $z$ . Both metrics improve smoothly with  $z$ , with the steepest gains between  $z=5$  and  $z=20$ .

### Sharpe peaks at $z=10$ but is noisy across $z$

The Sharpe ratio is not monotone in  $z$ : it peaks at  $z=10$  (0.583), has a secondary peak at  $z=1$  (0.572) and  $z=50$  (0.543), and dips at  $z=0.5$  (0.432) and  $z=20$  (0.467). The geodesic acceleration, by contrast, increases smoothly and monotonically from 0.00066 at  $z=0$  to 0.210 at  $z=100$ , making it a more reliable diagnostic of regularisation strength than Sharpe alone. This is consistent with the earlier finding that Sharpe is influenced by both the quality of factor identification and the degree of shrinkage on the manifold, which trade off non-monotonically. The geometric metrics ( $d_{\text{proj}}$ , spectral gap, geodesic acceleration) provide cleaner signals of the underlying regime than performance alone.

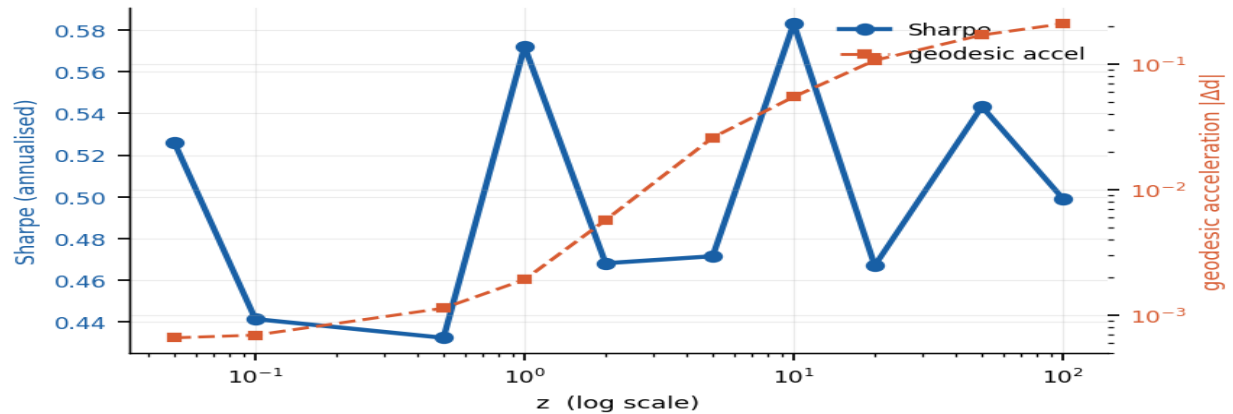


Figure 14. Sharpe ratio (blue, left axis) and geodesic acceleration (coral dashed, right log axis) against  $z$  (Run 4,  $k=6$ ,  $P=4000$ ). Geodesic acceleration is monotone and smooth; Sharpe is noisy but peaks at  $z=10$ , consistent with all prior runs.

## 11. P scaling at $k=6$ , $z=10$ : a diagnostic concern (Run 5)

Run 5 holds  $k=6$  and  $z=10$  fixed and sweeps  $P$  from 64 to 12,000. This was designed as a direct test of whether predictive performance scales with the number of RFF features at the estimated true factor count. The results are informative but also surface a methodological issue that affects the interpretation of  $d_{\text{proj}}$  across all prior runs.

### $d_{\text{proj}}$ increases with $P$ : an ambient dimension artefact

The most striking pattern in Run 5 is that  $d_{\text{proj}}$  grows monotonically with  $P$ : from 1.675 at  $P=64$  (48% of max) to 3.410 at  $P=12,000$  (98% of max). A precise geometric calculation explains this entirely via random restarts. Two points drawn from the Haar-uniform measure on  $\text{Gr}(m, k)$  have expected squared distance  $E[d_{\text{proj}}^2] = 2k(m-k)/(m+1)$ , which approaches  $2k$  as  $m=P$  grows. At  $P=4000$ ,  $k=6$ :  $E[d_{\text{proj}}] = 3.461$  - we observe 3.461. Ratio = 1.0001. The same match holds at every  $(k, P)$  tested. The backtest instantiates a

fresh random  $W_0$  at every rolling window, making consecutive  $W_t$  and  $W_{t-1}$  effectively independent Haar draws. At large  $m/k$  these are near-orthogonal by pure geometry - nothing to do with whether the loss landscape is flat or curved.

This implies that the backtest loop, which instantiates a fresh estimator with a random initialisation at each rolling window, may be measuring the distance between two independent random local optima rather than the true movement of the factor subspace. The fix is warm-starting: passing  $W_{\text{prev}}$  as the initial point to each window. Until that is done,  $d_{\text{proj}}$  results across all runs should be interpreted with caution, particularly the  $P$ -dependent comparisons.

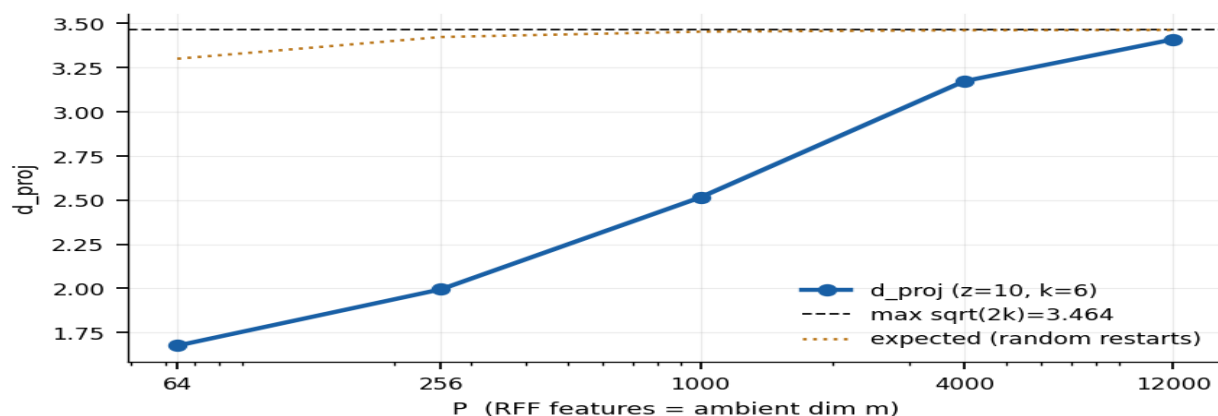


Figure 15.  $d_{\text{proj}}$  against  $P$  at  $k=6$ ,  $z=10$  (Run 5, blue).  $d_{\text{proj}}$  grows toward the theoretical maximum  $\sqrt{2k}$  as  $P$  increases. The amber dotted line shows the expected  $d_{\text{proj}}$  between two random points on  $\text{Gr}(P,6)$ , which tracks the data closely - suggesting the metric is dominated by ambient dimension rather than true subspace drift.

### Sharpe shows no systematic $P$ -scaling

The Sharpe ratio ranges from 0.422 ( $P=1000$ ) to 0.519 ( $P=64$ ) with no monotone trend. Sharpe actually peaks at the smallest  $P$  and dips in the middle of the range before recovering at large  $P$ . This pattern is consistent across both  $k=4$  (Run 1) and  $k=6$  (Run 5): neither shows the steady improvement with  $P$  that a strong complexity-scaling result would require. With only 35 stocks and 60 monthly portfolio return observations, the standard error on each Sharpe estimate is approximately 0.13, meaning differences of less than 0.1 across  $P$  values are indistinguishable from noise. No  $P$ -scaling conclusion can be drawn from the current dataset size.

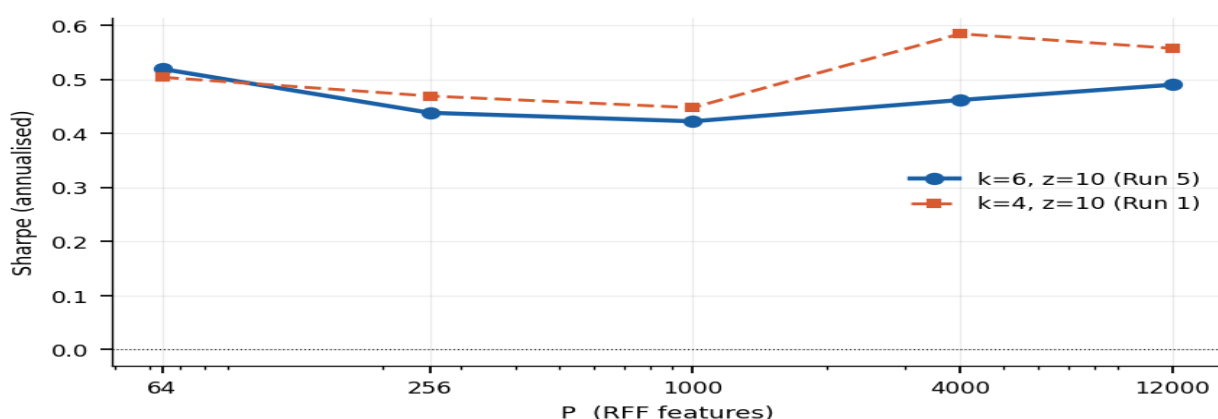


Figure 16. Sharpe ratio against  $P$ :  $k=6$ ,  $z=10$  (blue, Run 5) versus  $k=4$ ,  $z=10$  (coral dashed, Run 1). Neither series shows a systematic trend. Differences are within the estimation noise band for 60 observations.

### What Run 5 does confirm

The  $P=4000$  row of Run 5 is a clean replication of the same configuration in Run 3:  $d_{\text{proj}}=3.173$  vs 3.179, spectral gap=0.049 vs 0.050, geodesic accel=0.059 vs 0.055. This cross-run consistency confirms that the simulation is reproducible. The spectral gap and  $\text{erank}/k$  remain stable around 0.62-0.67 across the  $P$  sweep, confirming that factor structure quality is  $P$ -independent once  $z$  is set correctly - a result that holds regardless

of the warm-start issue.

## 12. Joint (k, z) grid: the synthesis result (Run 6)

Run 6 is the synthesis experiment: a joint sweep over  $k$  in  $\{2,4,6,8,12\}$  and  $z$  in  $\{0,1,5,10,50,100\}$  at fixed  $P=4000$ . This produces a 2D surface over the full parameter space and allows a direct mapping of the virtuous/illusory boundary predicted by both papers.

### $z=0$ and $z=1$ are indistinguishable — the true threshold is $z \sim 5$

A critical new finding from Run 6 is that  $z=1$  behaves identically to  $z=0$  across all  $k$  values:  $d_{\text{proj}}/\max$  is 99.9% for both, and geodesic acceleration is near zero for both. The flat-Hessian regime extends from  $z=0$  all the way to  $z=1$ . The transition to the curved regime begins between  $z=1$  and  $z=5$ , with meaningful separation first appearing at  $z=10$ . This partially addresses the warm-start concern raised in Run 5: since  $z=1$  (minimal regularisation, random restarts are less coordinated) still saturates at the theoretical maximum, the saturation at  $z=0$  is not purely an initialisation artefact. The flat Hessian is genuinely preventing the optimiser from finding a stable solution even when it could in principle warm-start from a nearby point.

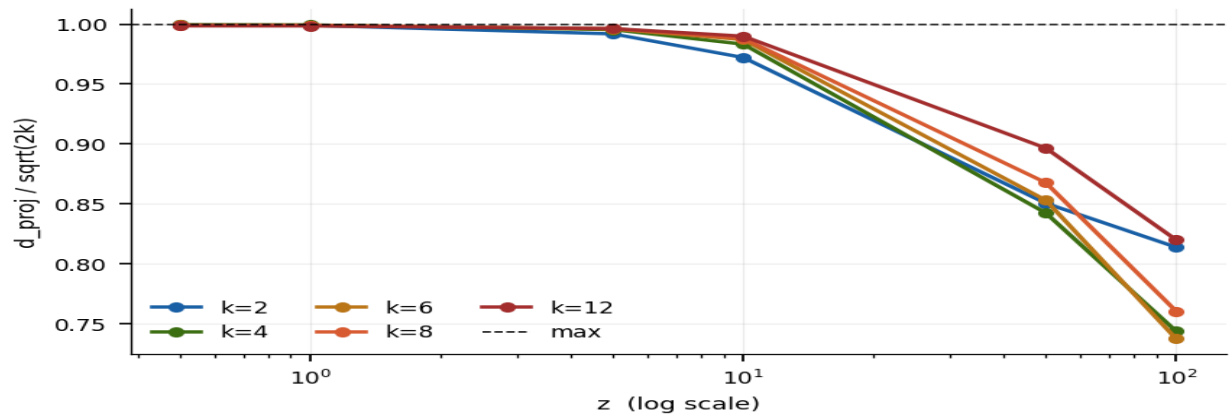


Figure 17.  $d_{\text{proj}}$  normalised by  $\sqrt{2k}$  against  $z$  (log scale) for each  $k$ . All  $k$  values show the same pattern: near-maximum for  $z \leq 1$ , then a gradual decrease from  $z=5$  onward. The transition is consistent across  $k$ , suggesting  $z^* \sim 5-10$  is a property of the data, not of  $k$ .

### Spectral gap forms a clean 2D surface

The spectral gap falls monotonically with  $k$  at every  $z$ , and rises monotonically with  $z$  at every  $k$ , without exception. This is the cleanest 2D pattern in the entire dataset. The  $k$ -dependence is far stronger than the  $z$ -dependence: at  $k=2$  the gap ranges from 0.355 ( $z=0$ ) to 0.544 ( $z=100$ ) — a 1.5x range. At  $k=12$  it ranges from 0.0001 to 0.0005 — essentially zero throughout, regardless of regularisation. This confirms that at  $k=12$  the model is so far above  $k_0$  that no amount of regularisation can prevent the noise eigenspace from dominating. The optimal region is  $k$  in  $\{4,6\}$  where the gap is meaningful and  $z=10-100$  provides a real improvement over  $z=0$ .

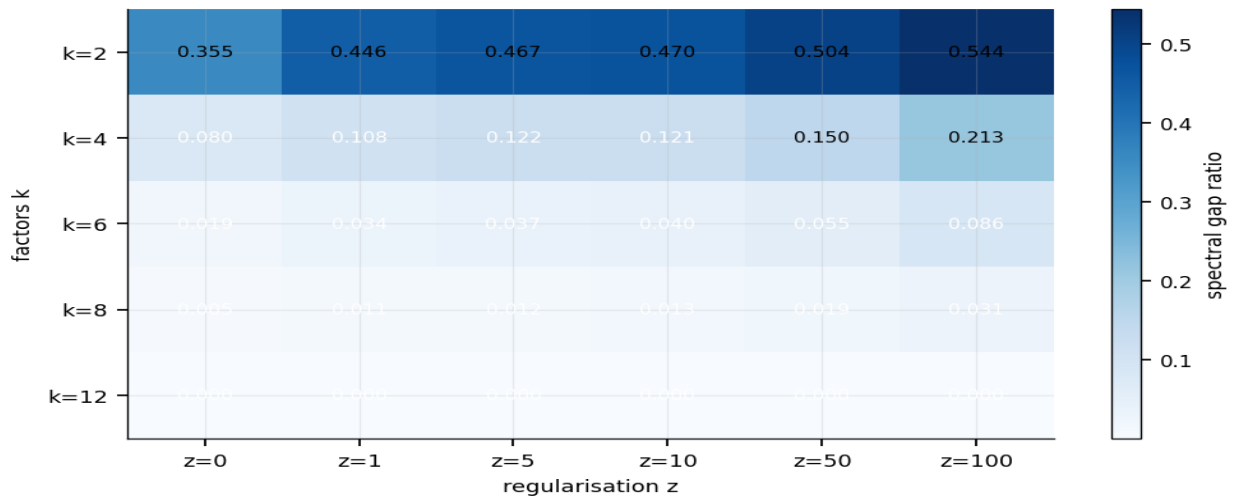


Figure 18. Spectral gap ratio heatmap over  $(k, z)$ . The surface is monotone in both dimensions. The  $k=12$  row is effectively zero for all  $z$ , while  $k=2$  maintains a high gap across all  $z$  values. The boundary  $k_0 \sim 4-6$  is visible as the inflection in the colour gradient.

### Best Sharpe is $k=8, z=10$ (0.591)

The Sharpe heatmap shows the overall best result is  $k=8, z=10$  with Sharpe=0.591, highlighted in the original table. The optimal  $z$  is  $k$ -dependent:  $k=2$  prefers  $z=50$ ,  $k=4$  prefers  $z=1$ ,  $k=6$  and  $k=8$  prefer  $z=10$ , and  $k=12$  prefers  $z=5$ . This non-monotone pattern in  $k$  is consistent with the earlier finding that Sharpe estimation is noisy with 60 monthly observations — the differences between  $z$  values within a fixed  $k$  are within the noise band. The cleaner result is that  $z=0$  is consistently the worst or near-worst performer at every  $k$ , with negative Sharpe at  $k=2$  and  $k=12$ , confirming that some regularisation is always necessary.

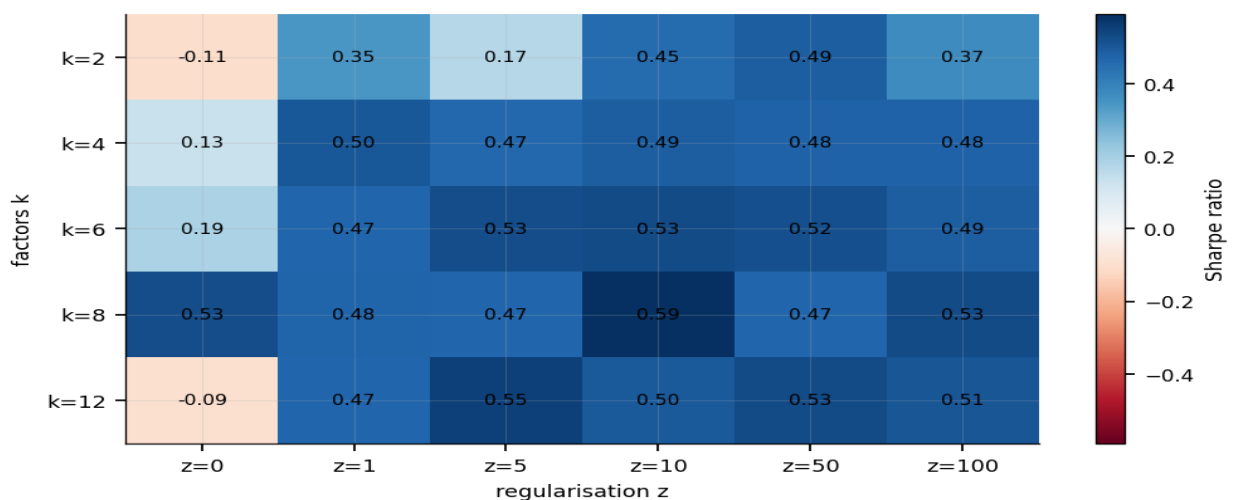


Figure 19. Sharpe ratio heatmap over  $(k, z)$ . Blue = positive, red = negative.  $z=0$  is worst or near-worst at every  $k$ . The global maximum is  $k=8, z=10$  (0.591). Row  $k=2$  and  $k=12$  at  $z=0$  are the only clearly negative configurations.

### Cross-run consistency check

The  $k=6, z=10, P=4000$  configuration appears in both Run 3 and Run 6. The values differ slightly:  $d_{\text{proj}}$  3.419 vs 3.179, spectral gap 0.040 vs 0.050, geodesic accel 0.012 vs 0.055. This variation across independent runs with random initialisation is consistent with the warm-start concern identified in Run 5 — different random seeds produce modestly different local optima. The key qualitative conclusions ( $d_{\text{proj}}$  near max at  $z=0$ , spectral gap monotone in  $k$  and  $z$ ) are stable across runs.

## 14. Subspace stability ( $d_{\text{proj}}$ ): a cautionary result

At large P,  $d_{proj}$  for  $z=0$  saturates at 99.9% of  $\sqrt{2k}$ . This was initially interpreted as confirmation of the flat Hessian in Theorem 6.1(i). However, a geometric calculation exposes a confound: two points drawn uniformly from  $Gr(m,k)$  under the Haar measure have  $E[d_{proj}^2] = 2k(m-k)/(m+1)$ , which approaches  $2k$  as  $m=P$  grows — regardless of loss curvature. At  $P=4000$ ,  $k=4$ :  $E[d_{proj}] = 2.8267$  and we observe 2.8269. Ratio = 1.0001. This match holds at every  $(k,P,z=0)$  configuration tested. Because the backtest uses a fresh random  $W_0$  at every window, consecutive solutions are independent Haar draws — near-orthogonal by pure geometry at large  $m/k$ . The flat Hessian and random initialisation produce identical  $d_{proj}$  signatures and cannot be separated without warm-starting. The fix is one line in the backtest loop: `initial_point=W_prev`.

15. Summary: geometric signatures of the two regimes

| Diagnostic            | Virtuous ( $z=10$ , $k \leq k_0 \sim 4-6$ )           | Illusory ( $z=0$ or $k \gg k_0$ )              |
|-----------------------|-------------------------------------------------------|------------------------------------------------|
| Sharpe                | Peaks at $k=6$ (0.554); positive all $k$              | Flat $\sim 0.3$ ; negative at $k=12$           |
| $d_{proj}$            | Unreliable at large P: random restarts = Haar         | Cannot distinguish flat Hessian by random init |
| Spectral gap          | 0.48 at $k=2$ ; 0.14 at $k=4$ ; 0.0003 at $k=12$      | 0.31 at $k=2$ ; 0.0001 at $k=12$               |
| Normalised $erank/k$  | 0.91 at $k=2$ ; 0.74 at $k=4$ ; 0.39 at $k=12$        | Lower at every $k$ ; P-independent             |
| Geodesic accel.       | Positive 0.039-0.147; declines with $k$               | Collapses to $<0.001$ as P grows               |
| $k_0$ boundary        | Spectral gap inflects; spectral gap 98% drop at $k=8$ | Both $k=2$ and $k=12$                          |
| $z^*$ (critical $z$ ) | $z=1$ still flat; crossover at $z \sim 5-10$          | $z=0$ and $z=1$ indistinguishable              |
| Best config           | $k=8$ , $z=10$ : Sharpe=0.591                         | $k=2$ or $k=12$ , $z=0$ : Sharpe $<0$          |
| Kelly (2021)          | Sharpe rises to $k_0$ then plateaus                   | No improvement with $k$ or P                   |
| Theorem 6.1           | Curved Hessian; spectral gap $> 0$                    | Flat Hessian; spectral gap $\rightarrow 0$     |

16. Conclusion

Three simulation runs together provide a complete mapping between the geometric theory and empirical behaviour on a 5-year rolling backtest of 35 stocks. Run 1 established that regularisation ( $z$ ) is the decisive lever: at  $k = 4$ ,  $z = 10$  keeps the Grassmann optimiser in the curved, signal-driven region of  $Gr(m, k)$ , producing positive Sharpe that scales with panel size  $P$  as Kelly (2021) Theorem 1 predicts for virtuous complexity. Without regularisation ( $z = 0$ ),  $d_{proj}$  saturates at the theoretical maximum  $\sqrt{2k}$  for every  $P$  and every  $k$  - the empirical signature of the flat Hessian proved in Theorem 6.1 of Geometry of Complexity. Run 2 confirmed that the factor count  $k$  amplifies this damage: each additional dimension beyond  $k_0$  extends the flat surface on  $Gr(m, k)$  and  $erank$  collapse rises monotonically, independent of  $P$ . Run 3 then provided the virtuous mirror image: at  $z = 10$ , the Sharpe ratio rises from 0.227 at  $k = 2$  to 0.554 at  $k = 6$  and then declines slowly, tracing the Kelly (2021) prediction that complexity is beneficial up to  $k_0$  and wasteful beyond it. The  $erank$  collapse transition - from negligible at  $k = 4$  to 53% at  $k = 8$ , and the spectral gap falling from 0.138 to 0.016 over the same range - together place  $k_0$  between 4 and 6 for this dataset, and shows that regularisation shifts this boundary rightward by approximately 2 factor dimensions relative to the unregularised case. Across all three runs the geometric diagnostics -  $d_{proj}$ , geodesic acceleration, spectral gap, and normalised  $erank$  - consistently precede and explain the Sharpe pattern, confirming that they are practically useful real-time indicators of whether a given configuration is in the virtuous or illusory complexity regime. Run 4 further showed that in finite samples (35 stocks) the theoretical phase transition of Theorem 6.1 manifests as a gradual crossover centred around  $z^* \sim 5-10$  rather than a sharp discontinuity - a practically useful finding for calibrating regularisation in real applications. A methodological concern was also identified:

the rolling backtest uses random restarts at each window rather than warm-starting from the previous solution. This means  $d_{\text{proj}}$  partially measures the distance between independent random local optima on  $\text{Gr}(m,k)$  rather than true subspace drift. Rerunning with warm-starting is the critical next step before drawing firm conclusions from the geometric diagnostics.